* Linear Optimization [Assignment 5]

Name : Sagar Parajuli , Roll no: CS26BTKMUJJ003

[References taken from Sundar Vishwanath lectures 9, 15 & 16.]

---

* Note on proof of primal optimal is equal to dual optimal.

Let us consider a primal problem in standard maximization form:

$$P: \quad \max \; c^T x \quad \text{s.t.} \; Ax \leq b, \; x \geq 0$$

where, $x \in \mathbb{R}^n$ is decision variables.

$A \in \mathbb{R}^{m \times n}$ is the constraint matrix

$b \in \mathbb{R}^m$ is the RHS vector

$c \in \mathbb{R}^n$ contains objective coefficients

Now, we can introduce a dual variable y_i for every primal constraint. Since the primal has m constraints, the dual has m variables. The corresponding dual is:

$$D: \quad \min \; b^T y \quad \text{s.t.} \; A^T y \geq c, \; y \geq 0$$

We can see this in expanded form as:

let P:

$$\begin{array}{lll} \max & c_1 x_1 + c_2 x_2 + \dots + c_n x_n & \\ \text{s.t.} & a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n \leq b_1 & \; y_1 \\ & a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n \leq b_2 & \; y_2 \\ & \vdots & \\ & a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n \leq b_m & \; y_m \end{array}$$

Suppose we are able to determine non-negative multipliers $y_1, y_2, \dots y_m$ such that, when the equations are multiplied by respective multipliers & added, we get objective function on the LHS, That is:

$$a_{11}y_1 + a_{21}y_2 + \dots + a_{m1}y_m = c_1$$
$$a_{12}y_1 + a_{22}y_2 + \dots + a_{m2}y_m = c_2$$
$$\vdots \qquad \vdots \qquad \vdots$$
$$a_{1n}y_1 + a_{2n}y_2 + \dots + a_{mn}y_m = c_n$$
$$y_i \geq 0, \quad \forall i = 1, \dots m$$

Then the sum of RHS i.e. $b_1y_1 + b_2y_2 + \dots + b_my_m$ gives an upper bound for the objective function of the primal.

Thm. The dual can be written as:

$$\min \; b_1y_1 + b_2y_2 + \dots + b_my_m$$
$$\text{s.t.} \quad a_{11}y_1 + a_{21}y_2 + \dots + a_{m1}y_m = c_1$$
$$a_{12}y_1 + a_{22}y_2 + \dots + a_{m2}y_m = c_2$$
$$\vdots$$
$$a_{1n}y_1 + a_{2n}y_2 + \dots + a_{mn}y_m = c_n$$
$$y_i \geq 0 \quad \forall i = 1, \dots m$$

* The duality theorem states the following:
  - if P is feasible & has a finite maximum, then D is feasible & the two optimum values coincide
  - if P is feasible & D is feasible then D is unbounded.
  - if P is feasible & unbounded then D is infeasible.

* Weak duality theorem: The primal & dual bound each other
  ∋ for any primal feasible x & dual feasible y
  $$c^Tx \leq b^Ty$$

<u>Proof:</u> Primal feasibility gives
$$Ax \leq b, \quad x \geq 0$$
& Dual feasibility gives
$$A^Ty \geq c, \quad y \geq 0$$

Multiply $Ax \leq b$ by y^T, since $y \geq 0$

$$y^T Ax \leq y^T b$$

Using the identity $y^T Ax = x^T A^T y$

$$x^T A^T y \leq b^T y$$

Since $A^T y \geq c$ & $x \geq 0$

$$c^T x \leq x^T A^T y$$

Combining the inequalities:

$$c^T x \leq x^T A^T y \leq b^T y$$

$$\therefore \quad c^T x \leq b^T y$$

Proved

⇒ Geometrically, for a maximization primal, every feasible solution provides a lower bound on the maximum attainable value. For a minimization dual, every feasible solution provides an upper bound on the minimum attainable value.

i.e. if Z_P^* & Z_D^* denote the primal & dual optimal values, then,

$$Z^*_P \leq Z^*_D$$

The primal cannot have an optimal value greater than dual's.

To establish that the gap between primal & dual's optimal is zero, i.e.

$$Z^*_P - Z^*_D \leq 0$$

or, $Z^*_D - Z^*_P \geq 0$

For the dual & primal optimal to be equal we must establish $Z^*_D - Z^*_P = 0$

i.e. $Z^*_D = Z^*_P$.

Thm, we introduce strong duality.

* Strong duality theorem: The primal optimal value equals the dual optimal value: For a primal-dual pair, if the primal has a finite optimal solution, then the dual also has an optimal solution & :

$$z_P^* = z_D^*$$

Conversely, if the dual has a finite optimal solution, the primal also has the same optimal.

Proof: From weak duality, we know

$$z_P^* \leq z_D^*$$

To establish strong duality, we must prove that there exists optimal feasible solutions x^* & y^* for which

$$c^T x^* = b^T y^*$$

Then,

$$z_P^* = c^T x^* = b^T y^* = z_D^*$$

$$\therefore z_P^* = z_D^*$$

To obtain x^* & y^*, we can use simplex optimality.

For the primal

$$\max \{ c^T x : Ax \leq b, x \geq 0 \}$$

The simplex optimality theorem guarantees, when a finite optimum exists, optimal solutions, x^* & y^*, satisfy

$$Ax^* \leq b, \quad x^* \geq 0$$

$$A^T y^* \geq c, \quad y^* \geq 0$$

The optimality conditions also give complementary slackness

$$y_i^* (b_i - a_i^T x^*) = 0, \quad \forall i \quad - (i)$$

$$x_j^* ((A^T y^*)_j - c_j) = 0, \quad \forall j \quad - (ii)$$

multiply & sum set of (i):

$$(y^*)^T(b - Ax^*) = 0$$

i.e. $$b^T y^* = (y^*)^T Ax^*$$

Similarly for (ii)

$$(x^*)^T(A^T y^* - c) = 0$$

i.e. $$c^T x^* = (x^*)^T A^T y^*$$

Since,

$$(y^*)^T Ax^* = (x^*)^T A^T y^*$$ [Received from complementary slackness]

we get,

$$b^T y^* = c^T x^*$$

& since x^* & y^* are both optimal solution

$$z_p^* = c^T x^* = z_D^* = b^T y^*$$

$$\therefore z_p^* = z_D^*$$

which is the required proof for strong duality obtained using the complementary slackness & simplex optimality theorem.

Thus, it's proved that primal optimal is equal to dual optimal & the two optimal bound each other, i.e. the maximisation of primal & minimization of dual meet at the same optimal point.

For example:

Consider the primal:

max: $x_1 + x_2$

s.t. $x_1 + x_2 \leq 4$

$x_1 \leq 2$

$x_1, x_2 \geq 0$

It's dual can be written as:

min: $4y_2 + 2y_2$

s.t. $y_1 + y_2 \geq 1$

$y_1 \geq 1$

$y_1, y_2 \geq 0$

For the primal the optimal value cannot exceed 4.

The $z_p^* = 4$ can be obtained at $(x_1, x_2) = (2, 2)$ which turns both the constraints into tight constraints.

i.e. $x_1 + x_2 = 4$

$x_1 = 2$

Thus, we reach an upper bound of maximum at point (2, 2) with optimum value of 4.

Similarly for the dual problem, at point (1, 0), both the constraints are tight i.e. $y_1 + y_2 = 1$ & $y_1 = 1$

& the minimum optimal value is obtained as $Z_D^* = 4$.

And, although the primal solution [2 2] & dual solution [1 0] are different vectors, their objective values coincide, which is what it means for the bounds to meet.

Thus, we prove that primal optimal is equal to dual optimal in the following steps:

We establish weak duality:

$$c^T x \leq x^T A^T y \leq b^T y$$

Then using complementary slackness

$$(y^*)^T (b - Ax^*) = 0$$

& $$(x^*)^T (A^T y^* - c) = 0$$

Then, we get $b^T y^* = (y^*)^T A x^*$

& $(x^*)^T A^T y^* = c^T x^*$

& since, $(y^*)^T A x^* = (x^*)^T A^T y^*$

we have, $b^T y^* = c^T x^*$

which is required proof of strong duality.